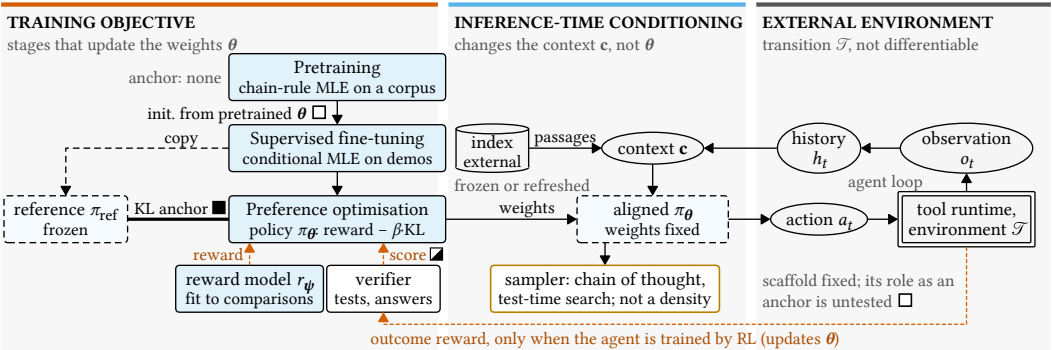
TRAINING OBJECTIVE
stages that update the weights θ
INFERENCE-TIME CONDITIONING
changes the context c, not θ
EXTERNAL ENVIRONMENT
transition 𝒯, not differentiable
anchor: none
Pretraining
chain-rule MLE on a corpus
init. from pretrained θ
copy
Supervised fine-tuning
conditional MLE on demos
index
external
passages
context c
history
h_t
observation
o_t
frozen or refreshed
agent loop
reference π_ref
frozen
KL anchor
Preference optimisation
policy π_θ; reward − βKL
weights
aligned π_θ
weights fixed
action a_t
tool runtime,
environment 𝒯
reward
score
reward model r_ψ
fit to comparisons
verifier
tests, answers
sampler: chain of thought,
test-time search; not a density
scaffold fixed; its role as an
anchor is untested
outcome reward, only when the agent is trained by RL (updates θ)
learned, updated
frozen
fixed operation
external store
environment
datum
data flow
training signal
evidence that an anchor matters:
established
observed
untested